

CONSTANT-CURRENT BATTERY CHARGER

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There are many ways of battery charging but constant-current charging, in particular, is a popular method for lead-acid and Ni-Cd batteries. In this circuit, the battery is charged with a constant current that is generally one-tenth of the battery capacity in ampere-hours. So for a 4.5Ah battery, constant charging current would be 450 mA.

This battery charger has the following features:

1. It can charge 6V, 9V and 12V batteries. Batteries rated at other voltages can be charged by changing the values of zener diodes ZD1 and ZD2.

2. Constant current can be set as per the battery capacity by using a potentiometer and multimeter in series with the battery.

3. Once the battery is fully charged, it will attain certain voltage level (e.g.

13.5-14.2V in the case of a 12V battery), give indication and the charger will switch off automatically. You need not remove the battery from the circuit.

4. If the battery is discharged below a limit, it will give deep-discharge indication.

5. Quiescent current is less than 5 mA and mostly due to zeners.

6. DC source voltage (V_{CC}) ranges from 9V to 24V.

7. The charger is short-circuit protected.

D1 is a low-forward-drop schottky diode SB560 having peak reverse voltage (PRV) of 60V at 5A or a 1N5822 diode having 40V PRV at 3A. Normally, the minimum DC source voltage should be 'D1 drop+Full charged battery voltage+ V_{DS} +R2 drop,' which is approximately 'Full charged battery voltage+5V.' For example, if we take full-charge voltage as 14V for a 12V battery, the source voltage should be

14+5=19V.

For the sake of simplicity, this constant-current battery charger circuit is divided into three sections: constant-current source, overcharge protection and deep-discharge protection sections.

The constant-current source is built around MOSFET T5, transistor T1, diodes D1 and D2, resistors R1, R2, R10 and R11, and potentiometer VR1. Diode D2 is a low-temperature-coefficient, highly stable reference diode LM236-5. LM236-5 can also be used with reduced operating temperature range of 0 to +70°C. Gate-source voltage (V_{GS}) of T5 is set by adjusting VR1 slightly above 4V. By setting V_{GS} , charging current can be fixed depending on the battery capacity. First, decide the charging current (one-tenth of the battery's Ah capacity) and then calculate the nearest standard value of R2 as follows:

$R2 = 0.7 / \text{Safe fault current}$

